📘 PART 2 — LEVEL 2 (Basic MCQs) ~120 Questions

🔶 SECTION 1: Amorphous & Crystalline Solids (Q1–15)

1. Which of the following is an amorphous solid?
   A) NaCl
   B) Glass
   C) Diamond
   D) Quartz
2. Crystalline solids have:
   A) Short-range order only
   B) Long-range order
   C) No order
   D) Random atomic positions
3. Which solid is isotropic in nature?
   A) Single crystal
   B) Amorphous solid
   C) BCC metal
   D) FCC metal
4. Which property is sharply defined in crystalline solids?
   A) Density
   B) Melting point
   C) Colour
   D) Solubility
5. Amorphous solids are also called:
   A) Supercooled liquids
   B) Polycrystals
   C) Single crystals
   D) Ionic solids
6. Which of the following solids has anisotropic properties?
   A) Glass
   B) NaCl crystal
   C) Rubber
   D) Plastic
7. Diamond is:
   A) Ionic solid
   B) Covalent solid
   C) Molecular solid
   D) Amorphous solid
8. Which solid shows plastic deformation easily?
   A) Metallic crystal
   B) Glass
   C) Quartz
   D) Diamond
9. Short-range order is exhibited by:
   A) Amorphous solids

B) Crystalline solids
C) Metallic solids
D) Ionic solids

10. Polycrystalline solids are made up of:
A) Single crystal
B) Many small crystals
C) Amorphous regions
D) None

11. Glass does not have:
A) Long-range order
B) Short-range order
C) Sharp melting point
D) Both A and C

12. Which of the following is a molecular solid?
A) Ice
B) NaCl
C) Diamond
D) Fe

13. Quartz is:
A) Amorphous
B) Crystalline
C) Metallic
D) Ionic

14. Ionic solids generally have:
A) Low melting points
B) High melting points
C) Softness
D) Plasticity

15. Diamond has coordination number:
A) 4
B) 6
C) 8
D) 12

🔶 SECTION 2: Crystal Lattices & Unit Cells (Q16–40)

16. Number of atoms in SC unit cell:
A) 1
B) 2
C) 4
D) 6

17. Number of atoms in BCC unit cell:
A) 1
B) 2
C) 4
D) 8

18. Number of atoms in FCC unit cell:
    A) 1
    B) 2
    C) 4
    D) 8
19. Coordination number in BCC:
    A) 6
    B) 8
    C) 12
    D) 4
20. Coordination number in FCC:
    A) 4
    B) 6
    C) 8
    D) 12
21. Coordination number in SC:
    A) 4
    B) 6
    C) 8
    D) 12
22. Edge length ‘a’ in SC in terms of radius r:
    A) a = r
    B) a = 2r
    C) a = 4r
    D) a = r/2
23. Edge length ‘a’ in BCC in terms of radius r:
    A) a = 2r
    B) a = 4r/√3
    C) a = r
    D) a = 3r
24. Edge length ‘a’ in FCC in terms of radius r:
    A) a = 2√2 r
    B) a = 4r
    C) a = 2r
    D) a = √3 r
25. Packing efficiency of SC:
    A) 52%
    B) 68%
    C) 74%
    D) 60%
26. Packing efficiency of BCC:
    A) 52%
    B) 68%
    C) 74%
    D) 60%
27. Packing efficiency of FCC:
    A) 52%
    B) 68%

C) 74%
D) 60%

28. Examples of FCC metals:
    A) Na, K
    B) Cu, Al, Au
    C) Fe, Cr
    D) Li, Cs
29. Examples of BCC metals:
    A) Na, K, Li
    B) Cu, Al, Au
    C) Fe, Cr, W
    D) Pb, Sn
30. Simple cubic unit cell has:
    A) 1 atom per unit cell
    B) 2 atoms per unit cell
    C) 4 atoms per unit cell
    D) 8 atoms per unit cell
31. Fractional contribution of corner atom:
    A) 1/8
    B) 1/4
    C) 1/2
    D) 1
32. Fractional contribution of face-centered atom:
    A) 1/8
    B) 1/4
    C) 1/2
    D) 1
33. Tetrahedral void is formed by:
    A) 3 atoms
    B) 4 atoms
    C) 6 atoms
    D) 8 atoms
34. Octahedral void is formed by:
    A) 4 atoms
    B) 6 atoms
    C) 8 atoms
    D) 12 atoms
35. Number of tetrahedral voids per atom in FCC:
    A) 1
    B) 2
    C) 4
    D) 6
36. Number of octahedral voids per atom in FCC:
    A) 1
    B) 2
    C) 3
    D) 4

37. Coordination number of tetrahedral void:
    A) 3
    B) 4
    C) 6
    D) 8
38. Coordination number of octahedral void:
    A) 3
    B) 4
    C) 6
    D) 8
39. HCP structure has packing efficiency:
    A) 52%
    B) 68%
    C) 74%
    D) 60%
40. In FCC, each atom is surrounded by:
    A) 6 nearest neighbours
    B) 8 nearest neighbours
    C) 12 nearest neighbours
    D) 4 nearest neighbours